

Liouvillian Geometry of Multidimensional Spectra: Pathway Transport and Observational Holonomy in Open Quantum Systems

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Liouville pathways provide the conceptual foundation for interpreting multidimensional spectroscopies and are commonly represented as independent contributions to the nonlinear response. In open quantum systems, however, the effective reduced Liouvillian need not preserve this observational pathway decomposition. Environmental interactions can transport amplitude between pathway sectors during the free-evolution intervals, generating measurable signatures in the nonlinear spectroscopic response. We develop a geometric framework in which pathway transport is governed by a Liouvillian connection, its associated curvature, and the resulting observational holonomy. The framework applies to open quantum systems in which the environment selects a pointer basis distinct from the observational basis used to construct the spectroscopic response. This basis incompatibility induces transport among Liouville pathways, generating characteristic spectral distortions and a nontrivial Liouvillian curvature. Using a Duhamel expansion of the Liouvillian propagator, we derive a reconstruction procedure that identifies the transport operators responsible for the observed redistribution of pathway weight, accurate throughout the full range of basis misalignment. This perspective reframes spectral features as determined not only by which pathways exist but by how amplitude is transported among them. Spectral distortions, peak shifts, and otherwise-forbidden pathway contributions become geometric signatures of a curved Liouville-space manifold rather than phenomenological broadening corrections, identifying pathway geometry as a complementary layer of organization in nonlinear spectroscopy.

I. INTRODUCTION

Multidimensional spectroscopies provide some of the most powerful tools available for probing quantum dynamics in condensed-phase and molecular systems. By correlating excitation, waiting, and detection intervals, techniques such as two-dimensional electronic spectroscopy (2DES), two-dimensional infrared spectroscopy (2DIR), and nonlinear magnetic resonance experiments resolve couplings, coherence transfer pathways, population relaxation, and environmental fluctuations that are inaccessible to linear response measurements^{1–6}. Central to the interpretation of these experiments is the Liouville-space pathway picture, introduced and systematized by Mukamel and co-workers^{1,7}, in which the measured signal is expressed as a sum of contributions generated by a sequence of field-matter interactions and subsequent propagation under an effective system Liouvillian. Double-sided Feynman diagrams provide a compact bookkeeping for these pathways and have become the standard language of nonlinear spectroscopy^{1–3,8,9}.

The pathway description has proven extraordinarily successful. It has been applied to study vibrational and electronic couplings^{8,10}, energy transfer and relaxation

dynamics in photosynthetic complexes^{9,11,12}, coherence and dephasing in molecular aggregates^{3,7}, and the interplay of vibrational and electronic coherence across a range of systems^{13,14}. Despite this success, the pathway picture rests on an implicit assumption: the basis used to define the pathways also provides an approximately independent and physically transparent representation of the dynamics between optical interactions.

In practice, quantum systems are never isolated. Interactions with the surrounding environment continuously reshape the evolving density matrix, generating relaxation, dephasing, coherence transfer, and population redistribution^{15–17}. As a result, the observed spectrum need not correspond precisely to the idealized pathway decomposition generated by a Hamiltonian model. When resolved in a fixed pathway basis, the resulting open-system dynamics may redistribute amplitude among nominally distinct pathway components, producing spectral distortions whose microscopic origin is often difficult to identify. Standard treatments address this through the secular Redfield approximation^{15,18}, which decouples population and coherence evolution and preserves the pathway structure, or through more complete treatments such as the hierarchical equations of motion (HEOM)^{19,20}, which capture non-Markovian effects but often obscure a simple pathway-based physical interpretation.

The failure of the observational basis to coincide with

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the bath-selected basis is not merely a technical complication; it has a precise physical meaning. Following Zurek's theory of environment-induced superselection (einselection)^{21–23}, the environment dynamically selects a preferred or pointer basis consisting of the states most robust under environmental monitoring. When the pointer basis and the Hamiltonian eigenbasis coincide, the pathway picture is exact and the secular approximation holds. When they do not, the environment acts in a basis incompatible with the pathway decomposition and continuously mixes pathways that the idealized model treats as independent. This basis incompatibility is the physical origin of pathway transport — and, as we show, of a non-trivial geometric structure on the space of spectroscopic observables.

This observation raises a natural inverse problem. Given a model Hamiltonian and its associated multidimensional spectrum, can one determine how environmental interactions modify the underlying pathway structure? Can the difference between an idealized spectrum and an observed spectrum be interpreted in terms of a transport process acting in Liouville space? Answering these questions would provide a route toward quantifying the dynamical influence of the environment directly from spectroscopic observables, without treating environmental effects solely through phenomenological broadening or relaxation parameters^{1,24}.

Geometric concepts have increasingly shaped our understanding of quantum dynamics and response. Berry phases and holonomies arising from cyclic evolution in parameter space leave observable signatures in spectroscopic signals^{25,26}, and quantum geometric tensors provide a unified framework connecting state distinguishability, adiabatic transport, and response^{27,28}. More recently, the stationary-state response of open quantum systems has been shown to admit a geometric decomposition into symmetric and antisymmetric sectors^{29,30}, with the antisymmetric sector governing nonreciprocal response and geometric work. The physical origin of this geometry is basis incompatibility between the Hamiltonian eigenbasis and the environment-selected pointer basis — the same incompatibility that, as we argue here, drives pathway transport in multidimensional spectroscopy.

The present work extends this framework into the domain of nonlinear spectroscopy. We argue that environmental dynamics continuously redistribute amplitude among Liouville pathways during the intervals separating successive field interactions, generating a transport process governed by a Liouvillian connection and its associated curvature. The resulting pathway transport leaves direct signatures in multidimensional spectra and gives rise to what we term *observational holonomy* — a geometric phase accumulated by the pathway basis as it is transported through Liouville space. Within a minimal rotated pointer-basis model, this transport is quantified by a single angle θ measuring the misalignment between the Hamiltonian eigenbasis and the environment-selected

pointer basis. We show that θ can be extracted directly from spectral observables, providing a route toward *geometric inverse spectroscopy*: reconstructing environmental pathway transport from multidimensional spectroscopic data.

The remainder of this paper is organized as follows. Section II establishes pathway transport as an intrinsic feature of open-system dynamics and identifies basis incompatibility as its physical origin. Section III develops the Liouvillian geometric framework — connections, curvature, holonomy, and observational holonomy — that gives pathway transport its geometric interpretation. Section IV introduces the minimal rotated pointer-basis model, derives the Liouvillian expansion, and constructs the pathway-transport matrix via the Duhamel identity. Section V presents the spectroscopic signatures of observational holonomy and demonstrates that a compact second-order transport expansion reproduces the exact response throughout the full range of basis misalignment. Section VI discusses the broader implications, including the contrast with microscopic approaches, the gauge invariance of the geometric response, and the program of geometric inverse spectroscopy.

II. PATHWAY TRANSPORT IN OPEN QUANTUM SYSTEMS

Conventional interpretations of multidimensional spectroscopic signals rest on the assignment of Liouville pathways generated by sequences of field-matter interactions. This framework has proven remarkably successful for describing coherence transfer, relaxation dynamics, and environmental fluctuations. Implicit in this construction, however, is the assumption that the pathway basis remains fixed throughout the experiment — that the basis used to define the pathways also provides an adequate representation of the dynamics between optical interactions.

Open quantum systems present a different situation. Pathways are not static objects but dynamical entities continuously reshaped by environmental interactions. The environment acts during every interval of free evolution, redistributing amplitude among nominally distinct pathways and generating coherence transfer and spectral distortions that cannot be understood solely in terms of the original pathway decomposition. The measured signal reflects the cumulative action of both optical and environmental transport.

To make this concrete, consider any system whose Hamiltonian eigenstates are delocalized superpositions of the states the environment monitors. During each free-evolution interval, the environment acts to suppress coherences in its preferred basis while the Hamiltonian generates coherences in its own eigenbasis. Neither wins outright. The result is a continuous redistribution of amplitude among the pathways that the idealized diagram treats as independent. The observed spectrum is not the

idealized spectrum; it is the idealized spectrum after being acted on by this transport process.

The dynamics are governed by a Liouvillian that separates naturally into three contributions,

$$\mathcal{L}(t) = \mathcal{L}_H + \mathcal{V}(t) + \mathcal{D}_{\text{bath}}, \quad (1)$$

where $\mathcal{L}_H \rho = -\frac{i}{\hbar}[H, \rho]$ generates coherent evolution, $\mathcal{V}(t)$ represents the external field interactions that generate the spectroscopic response, and $\mathcal{D}_{\text{bath}}$ collects the dissipative contributions from the environment. Pathway transport arises from the competition between these three contributions acting in generally incompatible directions in Liouville space. The discussion in this section is intentionally physical. A precise mathematical formulation of the manifold of physical models, the Liouvillian transport one-form, finite parametric transport, and the resulting definition of observational holonomy is developed in Appendix A.

The physical origin of that incompatibility is the existence of two distinct preferred bases. The *observational basis* is defined by the eigenstates of $\mathcal{L}_H$ — the Hamiltonian eigenbasis in which the pathway diagram is drawn. The *pointer basis* consists of the states most robust under environmental monitoring, selected dynamically through system–environment interactions following Zurek’s theory of environment-induced superselection^{21,23}. In Liouville space, these two structures are associated respectively with the eigenstructure of $\mathcal{L}_H$ and that selected by $\mathcal{D}_{\text{bath}}$.

We emphasize that this distinction is made at the level of the *reduced open-system description*. For the combined system and environment, the total dynamics may be represented as unitary evolution on an enlarged Hilbert space, in which case no fundamental separation between coherent and dissipative dynamics is required. After the environmental degrees of freedom are eliminated, however, the effective reduced generator contains coherent and dissipative structures that need not be simultaneously diagonalizable. The basis incompatibility discussed here therefore does not assert an absolute distinction between microscopic coherent and dissipative dynamics. Rather, it characterizes the reduced Liouvillian relative to the observational basis in which the nonlinear spectroscopic pathways are resolved.

Pathway transport emerges when these structures are incompatible,

$$[\mathcal{L}_H, \mathcal{D}_{\text{bath}}] \neq 0, \quad (2)$$

so that the Hamiltonian-defined pathway basis and the bath-selected pointer basis cannot be simultaneously diagonalized. An analogous incompatibility between field-induced and dissipative transport,

$$[\mathcal{V}(t), \mathcal{D}_{\text{bath}}] \neq 0, \quad (3)$$

further redistributes amplitude throughout the pathway network during the optical interactions themselves.

When both commutators vanish, the conventional pathway picture is recovered exactly.

Within this reduced description, the noncommutativity is objective: once the subsystem, effective Liouvillian, and observational pathway basis are specified, whether the corresponding dynamical sectors can be simultaneously diagonalized is a basis-independent statement about that reduced generator. A different microscopic system–bath representation may produce the same reduced transport structure.

Multidimensional spectroscopy thus becomes a problem of transport on a network of pathways. The objective is not to identify every microscopic environmental process individually, but to characterize how environmental interactions reshape the pathway structure itself. The relevant quantities are the transport processes induced within Liouville space, not the bath modes or spectral densities that drive them. This reframing provides the foundation for the geometric framework developed in the following section.

III. LIOUVILLIAN GEOMETRY OF PATHWAY TRANSPORT

Quantifying pathway transport requires specifying how Liouville-space pathway bases are compared as the system parameters vary — a problem that is naturally addressed by the language of connections and curvature. The objects being transported are not vectors in physical space but Liouvillian eigenoperators and the spectroscopic pathways they define. The mathematical foundations of this geometric construction are developed in Appendix A, where the manifold of physical models, the Liouvillian transport one-form, finite parametric transport, and the resulting observational holonomy are introduced from first principles.

A. Transport of Liouvillian Eigenoperators

Transport of the pathway basis through Liouville space arises whenever the Liouvillian eigenoperators vary with the control parameters. Consider a Liouvillian $\mathcal{L}(\lambda)$ depending on a set of external controls λ^μ , which may represent thermodynamic variables, external fields, pulse-sequence parameters, or properties of the environment. The corresponding Liouvillian eigenproblem is

$$\mathcal{L}(\lambda)\Phi_n(\lambda) = \Lambda_n(\lambda)\Phi_n(\lambda), \quad (4)$$

where Λ_n are the Liouvillian eigenvalues and Φ_n denote the associated right eigenoperators. For non-Hermitian Liouvillians it is necessary to introduce a biorthogonal set of left eigenoperators satisfying

$$\tilde{\Phi}_m^\dagger(\lambda)\mathcal{L}(\lambda) = \Lambda_m(\lambda)\tilde{\Phi}_m^\dagger(\lambda), \quad (5)$$

together with the normalization condition

$$\langle\langle\tilde{\Phi}_m|\Phi_n\rangle\rangle = \delta_{mn}, \quad (6)$$

where $\langle\langle A|B\rangle\rangle = \text{Tr}(A^\dagger B)$ denotes the Hilbert–Schmidt inner product.

As the control parameters are varied, both the eigenvalues and the eigenoperators may change. These two effects play fundamentally different roles. Variations of the eigenvalues modify relaxation rates, dephasing rates, and oscillation frequencies, but do not by themselves alter the underlying pathway structure. By contrast, variations of the eigenoperators change the basis in which the dynamics are represented and therefore induce transport of spectroscopic pathways through Liouville space.

Geometric transport is governed by the evolution of the eigenoperators, not the eigenvalues. A Liouvillian may exhibit strong parameter dependence through its eigenvalues while possessing no nontrivial pathway transport. Such behavior occurs, for example, in the conventional reduced pure-dephasing model, where the dephasing rate may vary with the control parameters while the Liouvillian eigenoperator structure remains unchanged. Here, “pure dephasing” refers to the standard reduced open-system description in which coherences decay without population transfer in the observational basis (e.g., Refs. 16,17). In that case the pathway basis is fixed and no geometric transport is generated.

Nontrivial geometric response emerges only when the eigenoperators themselves evolve with the control parameters. In that situation, spectroscopic pathways defined at one point in parameter space can no longer be identified unambiguously with pathways defined at another. Comparing pathway bases at neighboring points therefore requires a rule for transport between local Liouville-space frames. The Liouvillian connection specifies that transport law.

As shown in Appendix A, this transport law arises naturally from the parametric variation of the Liouvillian propagator, leading to an operator-valued transport one-form whose local representation is developed in the present section.

B. Liouvillian Connections

The natural measure of how the pathway basis changes under an infinitesimal displacement in parameter space is the derivative of the Liouvillian itself. We define the infinitesimal Liouvillian deformation by

$$\Gamma_\mu = \partial_\mu \mathcal{L}, \quad (7)$$

where $\partial_\mu \equiv \partial/\partial\lambda^\mu$. When $\Gamma_\mu = 0$, the Liouvillian is independent of λ^μ and no pathway transport is generated in that direction. When $\Gamma_\mu \neq 0$, the local structure of the Liouvillian changes with the controls, and the pathway basis is transported through Liouville space. The connection therefore measures the local rate of that transport.

The transport generated by Γ_μ is most naturally expressed in the local Liouvillian eigenbasis,

$$(\Gamma_\mu)_{mn} = \langle\langle\tilde{\Phi}_m|\Gamma_\mu|\Phi_n\rangle\rangle, \quad (8)$$

where Φ_n and $\tilde{\Phi}_n$ denote the right and left Liouvillian eigenoperators introduced in Eq. (4). The diagonal elements $(\Gamma_\mu)_{nn}$ govern the local response of individual pathways to variations of the control parameters. The off-diagonal elements $(\Gamma_\mu)_{m\neq n}$ couple distinct pathways and generate transport throughout the pathway network, providing a direct measure of pathway mixing in Liouville space.

The Liouvillian connection has a direct analogue in the geometric description of adiabatic quantum evolution. For a parameter-dependent Hamiltonian,

$$H(\lambda)|n(\lambda)\rangle = E_n(\lambda)|n(\lambda)\rangle, \quad (9)$$

transport of the eigenstates is governed by the Berry connection,

$$A_{\mu,n} = i\langle n|\partial_\mu n\rangle. \quad (10)$$

The Berry connection governs the transport of quantum states under variations of external controls. The Liouvillian connection introduced in Eq. (7) plays the corresponding role in Liouville space. The transported objects are no longer wavefunctions but Liouvillian eigenoperators and the spectroscopic pathways they define.

The distinction is important. Berry geometry describes transport on Hilbert space and is therefore restricted to coherent state evolution. Liouvillian geometry, by contrast, is defined on the space of dynamical generators and incorporates dissipation, decoherence, and environmental transport on equal footing. In the special case of purely unitary dynamics,

$$\mathcal{L}\rho = -\frac{i}{\hbar}[H, \rho], \quad (11)$$

the Liouvillian geometry reduces to the corresponding geometry of the underlying Hamiltonian. The dissipative sector introduces transport processes with no direct analogue in conventional Berry theory. In the observational pathway basis, purely Hamiltonian propagation preserves the decomposition when the corresponding pathway sectors are invariant under the coherent generator. Dissipative terms that are not diagonal in this same decomposition introduce off-diagonal couplings between those sectors and hence generate pathway transport.

C. Curvature, Holonomy, and Observation

The geometric quantities introduced below admit a more general differential-geometric interpretation developed in Appendix A. There, the Liouvillian connection is shown to arise from an operator-valued transport one-form whose path-ordered integration generates

finite parametric transport, providing the mathematical foundation for the notions of holonomy and observational completion employed here.

The distinction between state variables and transport processes is familiar from thermodynamics. Internal energy is a state variable: the change between two equilibrium states depends only on the endpoints. Work, by contrast, depends on the path taken through state space, precisely because it characterizes transport rather than a property of the state itself. The same distinction underlies pathway transport in Liouville space: the pathway basis at a given point in parameter space is a local quantity, but how it was transported there depends on the route.

The degree to which successive transport operations fail to commute is quantified by the Liouvillian curvature,

$$\mathcal{K}_{\mu\nu} = \partial_\mu \Gamma_\nu - \partial_\nu \Gamma_\mu + [\Gamma_\mu, \Gamma_\nu]. \quad (12)$$

The first two terms measure how the transport generators vary across parameter space. The commutator term $[\Gamma_\mu, \Gamma_\nu]$ arises because the connection operators themselves need not commute: transporting the pathway basis first along λ^μ then along λ^ν need not produce the same result as the reverse sequence. In open quantum systems this noncommutativity has a direct physical origin — coherent, optical, and dissipative dynamics define competing transport directions in Liouville space, and the curvature quantifies the extent to which those directions are incompatible.

Curvature characterizes the local structure of pathway transport, while its global consequences emerge through transport around finite closed loops in parameter space. Transport around a closed contour C need not return the pathway basis to its original configuration. The resulting net transformation is the holonomy,

$$\mathcal{U}_C = \mathcal{P} \exp \left[\oint_C \Gamma_\mu d\lambda^\mu \right], \quad (13)$$

where $\mathcal{P}$ denotes path ordering — the instruction that the exponential is to be evaluated with transport operations applied in the order dictated by the contour, since Γ_μ operators at different points need not commute. The holonomy captures the cumulative effect of Liouville-space transport around the loop and is the global manifestation of the underlying curvature.

Spectroscopy, however, accesses only a projection of this transport. The measured signal,

$$S(t) = \text{Tr}[O\rho(t)], \quad (14)$$

selects a specific component of the underlying Liouville-space dynamics through the observable O . Observation therefore does not reveal the full holonomy $\mathcal{U}_C$ but only its projection onto the measurement basis³¹. We refer to this projected quantity as the observational holonomy.

This distinction is important. The transport geometry is a property of the underlying Liouvillian dynamics, whereas the observational holonomy is a property

of those dynamics viewed through a particular observable³¹. Different observables define different projections of the same transport process and need not reveal the same geometric structure. Two experiments probing the same open quantum system may therefore exhibit different observational holonomies despite being governed by the same underlying Liouvillian transport.

The measured spectrum thus resides not directly in the transport manifold itself but in an observational manifold generated by the chosen set of observables. The geometry accessible to experiment is therefore jointly determined by the dynamics and by the measurement channel through which those dynamics are interrogated. Reconstructing transport geometry from measured spectra is therefore simultaneously a problem of dynamical evolution and observable projection, providing the inverse problem that the remainder of this paper addresses.

IV. MINIMAL MODEL FOR OBSERVATIONAL HOLONOMY

A. Pointer-Basis Misalignment

The geometric framework developed in Sec. III connects directly to results established for the stationary-state response of open quantum systems^{29,30}. There it was shown that when the Hamiltonian eigenbasis and the environment-selected pointer basis are misaligned by an angle θ , the stationary-state response tensor develops an antisymmetric sector — a curvature two-form whose physical origin is precisely the basis incompatibility quantified by θ . That work asked what thermodynamic work the resulting geometry performs over a quasistatic control cycle. The present work asks a different question: what does that same geometry do to a nonlinear optical experiment? The minimal model introduced below is the same model; the observable is different. Instead of work accumulated over a closed cycle in parameter space, we examine amplitude redistributed among spectroscopic pathways during the waiting intervals of a multidimensional experiment.

The system consists of a two-level Hamiltonian

$$H = \frac{\Delta}{2} \sigma_z, \quad (15)$$

whose eigenstates define the observational basis and the corresponding spectroscopic pathways. In the absence of environmental coupling, the pathway structure remains fixed and yields the conventional decomposition of the optical response.

The environment selects a preferred basis distinct from the observational basis. The two are related by the unitary transformation

$$|p_n\rangle = U(\theta)|e_n\rangle, \quad (16)$$

where $|e_n\rangle$ and $|p_n\rangle$ denote the observational and pointer states, respectively, and θ quantifies their misalignment.

Importantly, θ does not represent a physical rotation in time — it is a fixed structural parameter characterizing the relationship between the basis the experimenter uses to define pathways and the basis the environment selects through decoherence.

The physical content of θ is transparent. At $\theta = 0$, the pointer and observational bases coincide: the environment acts through pure dephasing in the observational basis, suppressing coherences while leaving populations unchanged, and no pathway transport is generated. As θ increases, the dissipator acquires components that couple populations and coherences, opening relaxation channels that redistribute amplitude among observational pathways. The angle θ therefore measures the degree to which environmental monitoring departs from pure dephasing — and equivalently, the degree to which pathway transport is activated. Recovering θ from spectroscopic data amounts to reconstructing the balance between dephasing, relaxation, and pathway mixing generated by the environment.

The basis mismatch becomes dynamically relevant through the Liouvillian governing the open-system evolution,

$$\dot{\rho} = \mathcal{L}(\boldsymbol{\lambda})\rho, \quad (17)$$

where $\boldsymbol{\lambda}$ denotes the control parameters governing the observational–pointer relationship. For the minimal model, $\boldsymbol{\lambda} = \theta$; more generally, the control manifold may include multiple environmental couplings, anisotropic relaxation channels, or other mechanisms that deform the Liouvillian structure away from the reference dynamics.

Expanding locally about the aligned reference point $\boldsymbol{\lambda} = 0$,

$$\mathcal{L}(\boldsymbol{\lambda}) = \mathcal{L}_0 + \sum_{\mu} \lambda^{\mu} \mathcal{X}_{\mu} + \frac{1}{2} \sum_{\mu\nu} \lambda^{\mu} \lambda^{\nu} \mathcal{Y}_{\mu\nu} + O(\lambda^3), \quad (18)$$

where

$$\mathcal{X}_{\mu} = \left. \frac{\partial \mathcal{L}}{\partial \lambda^{\mu}} \right|_{\boldsymbol{\lambda}=0}, \quad \mathcal{Y}_{\mu\nu} = \left. \frac{\partial^2 \mathcal{L}}{\partial \lambda^{\mu} \partial \lambda^{\nu}} \right|_{\boldsymbol{\lambda}=0}. \quad (19)$$

The operators $\mathcal{X}_{\mu}$ generate the leading transport of the pathway basis away from the reference dynamics, while $\mathcal{Y}_{\mu\nu}$ describe the nonlinear corrections. For the minimal model, the expansion reduces to

$$\mathcal{L}(\theta) = \mathcal{L}_0 + \theta \mathcal{X} + \frac{\theta^2}{2} \mathcal{Y} + O(\theta^3). \quad (20)$$

To quantify this transport, we begin with the conventional pathway decomposition

$$S^{(0)} = \sum_a S_a^{(0)}, \quad (21)$$

where a labels the zeroth-order Liouville pathways. In the aligned limit the pathways remain dynamically independent. When the Liouvillian is deformed, transport

between pathways is generated and the response becomes

$$S_a(\boldsymbol{\lambda}) = \sum_b T_{ab}(\boldsymbol{\lambda}) S_b^{(0)}, \quad (22)$$

where T_{ab} is the pathway-transport operator. The diagonal elements describe modifications of individual pathway weights, while the off-diagonal elements quantify transport between otherwise independent pathways. When $T_{ab} = \delta_{ab}$, each pathway evolves independently and the conventional decomposition is recovered.

To construct T_{ab} explicitly, we expand the Liouvillian propagator

$$U(\tau; \boldsymbol{\lambda}) = e^{\mathcal{L}(\boldsymbol{\lambda})\tau} \quad (23)$$

in powers of the deformation parameters. The Duhamel identity^{17,32,33} provides a natural route: it expresses the propagator of a perturbed generator as a series of integrals over the reference propagator, with each order capturing successive insertions of the deformation operator into the free evolution. Applying it to Eq. (20) yields

$$U(\tau; \boldsymbol{\lambda}) = U_0(\tau) + \sum_{\mu} \lambda^{\mu} U_{\mu}^{(1)}(\tau) + \frac{1}{2} \sum_{\mu\nu} \lambda^{\mu} \lambda^{\nu} U_{\mu\nu}^{(2)}(\tau) + O(\lambda^3), \quad (24)$$

where $U_0(\tau) = e^{\mathcal{L}_0\tau}$ is the reference propagator and

$$U_{\mu}^{(1)}(\tau) = \int_0^{\tau} ds e^{\mathcal{L}_0(\tau-s)} \mathcal{X}_{\mu} e^{\mathcal{L}_0 s}. \quad (25)$$

The first-order term describes a single insertion of the deformation $\mathcal{X}_{\mu}$ at time s , sandwiched between reference propagators before and after. The second-order contribution,

$$U_{\mu\nu}^{(2)}(\tau) = \int_0^{\tau} ds e^{\mathcal{L}_0(\tau-s)} \mathcal{Y}_{\mu\nu} e^{\mathcal{L}_0 s} + 2 \int_0^{\tau} ds \int_0^s ds' e^{\mathcal{L}_0(\tau-s)} \mathcal{X}_{\mu} e^{\mathcal{L}_0(s-s')} \mathcal{X}_{\nu} e^{\mathcal{L}_0 s'}, \quad (26)$$

contains two physically distinct terms: a single insertion of the second-order deformation $\mathcal{Y}_{\mu\nu}$, and a double insertion of the first-order deformation $\mathcal{X}_{\mu}$ at two successive times. The latter captures the cumulative effect of two pathway-transport events separated by free evolution under $\mathcal{L}_0$.

The quantities $U_{\mu}^{(1)}(\tau)$ and $U_{\mu\nu}^{(2)}(\tau)$ describe the first- and second-order corrections to the full Liouvillian propagator. To obtain the corresponding pathway transport coefficients, these propagators are projected onto the reference pathway basis defined by the zeroth-order Liouvillian eigenoperators. The resulting matrix elements define the transport tensors $T_{ab,\mu}^{(1)}$ and $T_{ab,\mu\nu}^{(2)}$ appearing below.

Projecting onto the zeroth-order pathway basis yields the transport matrix,

$$T_{ab}(\lambda) = \delta_{ab} + \sum_{\mu} \lambda^{\mu} T_{ab,\mu}^{(1)} + \frac{1}{2} \sum_{\mu\nu} \lambda^{\mu} \lambda^{\nu} T_{ab,\mu\nu}^{(2)} + O(\lambda^3), \quad (27)$$

where $T_{ab} = \delta_{ab}$ in the aligned limit, recovering the conventional pathway decomposition. The off-diagonal coefficients $T_{ab,\mu}^{(1)}$ and $T_{ab,\mu\nu}^{(2)}$ quantify pathway mixing at first and second order in the deformation. Specifically,

$$T_{ab,\mu}^{(1)} = \langle\langle \tilde{\Phi}_a^{(0)} | U_{\mu}^{(1)} | \Phi_b^{(0)} \rangle\rangle,$$

and similarly

$$T_{ab,\mu\nu}^{(2)} = \langle\langle \tilde{\Phi}_a^{(0)} | U_{\mu\nu}^{(2)} | \Phi_b^{(0)} \rangle\rangle,$$

where the superscript (0) denotes the eigenoperators of the reference Liouvillian. We also note that while the present derivation is carried out in the temporal representation using the propagator $U(\tau)$. Appendix A shows that the construction extends directly to the more general Liouvillian propagator $\mathcal{G}(\xi; \lambda)$, unifying the temporal exponential and frequency-domain resolvent representations within a common transport framework.

The second-order deformation tensors $\mathcal{Y}_{\mu\nu}$ should not be confused with the Liouvillian curvature introduced in Sec. III. The quantities $\mathcal{Y}_{\mu\nu}$ arise from the local Taylor expansion of the Liouvillian and describe nonlinear changes in the dynamical generator itself. The curvature, by contrast, is constructed from the Liouvillian connection and measures the noncommutativity of transport on the control manifold. Both derive from the same underlying Liouvillian geometry, but carry distinct physical content: $\mathcal{Y}_{\mu\nu}$ characterizes local deformations of the generator, while the curvature characterizes the geometry of transport those deformations induce.

Equation (27) provides an operational representation of observational holonomy. The spectroscopic inverse problem reduces to reconstructing a finite-dimensional transport matrix acting on the observable pathways — not optimizing agreement with the observed response, but inferring the transport processes acting between pathways. The deformation parameters λ^{μ} are determined by the local structure of the Liouvillian, not introduced as phenomenological fit parameters. What is extracted from experiment is therefore not a set of adjustable numbers but a physically interpretable description of how the environment redistributes amplitude among spectroscopic pathways.

The Duhamel expansion plays the role of ordinary perturbation theory in this reconstruction. The deformation operators $\mathcal{X}_{\mu}$ identify the directions in Liouvillian space responsible for pathway transport, while successive orders quantify their contribution to the observed response. Spectral deviations are decomposed into physically interpretable transport processes rather than phenomenological corrections. The remainder of this work examines

how this transport appears in nonlinear spectra and how θ can be extracted from experimental data.

V. SPECTROSCOPIC SIGNATURES OF OBSERVATIONAL HOLONOMY

The geometric framework developed above predicts that Liouvillian deformations generate transport between observational pathways. This transport modifies the nonlinear spectroscopic response and provides an observable manifestation of observational holonomy. To illustrate these effects, we return to the minimal two-level model introduced in Sec. IV A. Rather than analyzing the full two-dimensional spectrum, we focus on diagonal spectral slices, which retain the essential signatures of pathway transport while providing a compact representation of the response.

In practice, the mismatch between the observational pathway basis and the effective reduced Liouvillian is not observed directly but is inferred from systematic deviations between the measured spectrum and that predicted by a reference Liouvillian. Figure 2 provides a concrete illustration of this inverse problem. Beginning with a reference Liouvillian, the Duhamel transport expansion reconstructs the spectral distortions produced by an unknown basis mismatch. The recovered transport operators therefore provide a quantitative measure of the underlying pathway transport and, consequently, of the observational holonomy associated with the reduced dynamics.

A. Diagonal Spectral Response

Figure 1 shows the real and imaginary components of the diagonal response for increasing pointer-basis misalignment. The Hamiltonian and optical transition operators are held fixed throughout; consequently, all observed spectral changes arise from the environmentally induced transport of pathway weight.

As the pointer basis departs from the observational basis, both the real and imaginary components of the response evolve continuously and in a highly structured manner. The real component tracks the redistribution of spectral weight among the contributing pathways, while the imaginary component is particularly sensitive to the accompanying phase relationships. Basis misalignment does more than broaden or attenuate the response — it transports pathway weight through Liouville space, producing measurable deformations of the complex spectrum. These deformations are the direct spectroscopic manifestation of observational holonomy.

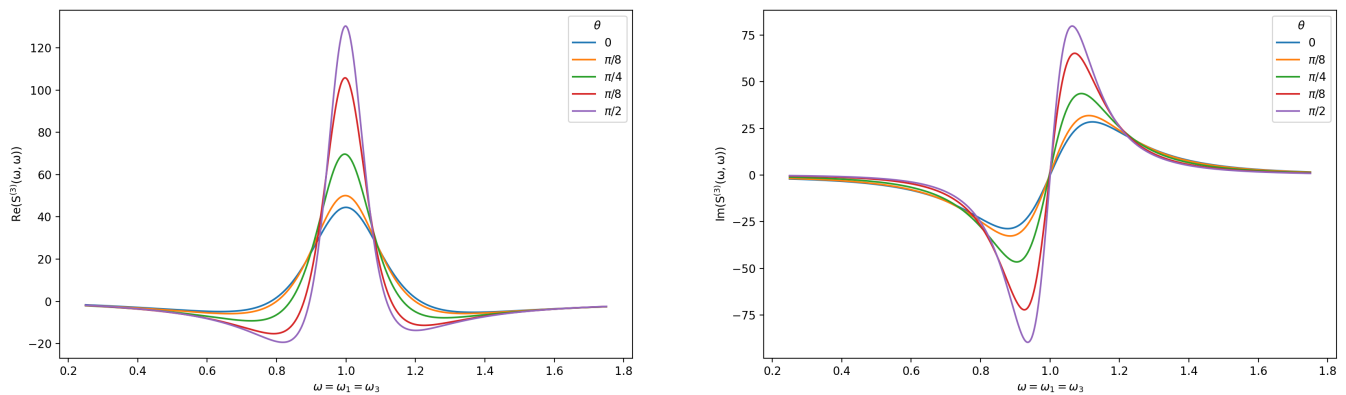


FIG. 1: Real (left) and imaginary (right) components of the diagonal spectral response for increasing pointer-basis misalignment θ . At $\theta = 0$, only dephasing contributions are present. For $\theta \neq 0$, basis misalignment mixes response pathways, producing continuous and structured deformations of the complex spectrum. These serve as benchmarks for the Duhamel reconstructions shown in Fig. 2.

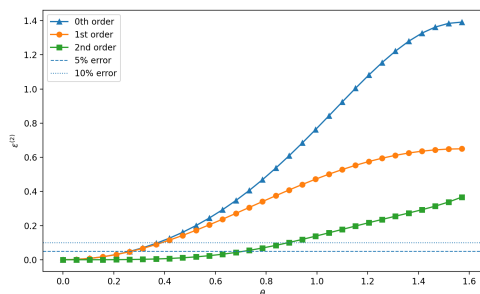


FIG. 2: Global reconstruction error $\epsilon^{(n)}$ for the first- and second-order Duhamel transport expansions as a function of pointer-basis misalignment θ . The zeroth-order curve compares the $\theta = 0$ signal to the exact result at $\theta \neq 0$. The second-order approximation remains accurate throughout the full range of misalignment, including at $\theta = \pi/2$ where the observational and pointer bases are maximally misaligned.

B. Accuracy of the Transport Expansion

Figure 2 shows that the transport expansion remains accurate throughout the full range of basis misalignment. The first-order approximation captures the leading effects of pathway transport; the second-order terms substantially improve the reconstruction and remain accurate even at $\theta = \pi/2$, where the observational and pointer bases are maximally misaligned.

This robustness is the key quantitative result of this section. Although the Duhamel expansion is derived from local Liouvillian deformations about $\theta = 0$, it remains accurate well beyond the regime where a weak-deformation approximation might be expected to hold. The objective is therefore not spectral reconstruction *per se*, but geometric reconstruction. Rather than merely re-

producing an observed multidimensional spectrum, the transport expansion reconstructs the underlying deformation of the reduced Liouvillian responsible for that spectrum. The dominant spectroscopic consequences of pathway transport are therefore captured by a compact set of transport coefficients, which quantify the underlying transport geometry.

Viewed in this way, Fig. 2 provides a concrete proof-of-principle for geometric inverse spectroscopy. Beginning with a reference Liouvillian, the Duhamel transport expansion reconstructs the spectral distortions generated by an unknown basis misalignment. The resulting transport coefficients do more than reproduce the observed spectrum: they recover the underlying pathway transport and thereby infer the deformation of the reduced Liouvillian responsible for the measured response. The objective is therefore not spectral reconstruction *per se*, but geometric reconstruction. The compact hierarchy of Duhamel transport coefficients thus provides a practical means of inferring the transport geometry—and the associated observational holonomy—from multidimensional spectroscopic measurements.

C. Interpretation

The Duhamel expansion does more than provide a reconstruction algorithm — it associates spectral changes with specific transport processes. Each order in the expansion corresponds to a physically distinct mechanism: first-order terms describe single pathway-mixing events, while second-order terms capture either nonlinear deformations of the generator or two successive mixing events separated by free evolution. Spectral deviations from the reference response are thereby decomposed into a hierarchy of transport processes rather than absorbed into phenomenological line-shape parameters.

This decomposition suggests a natural inverse problem.

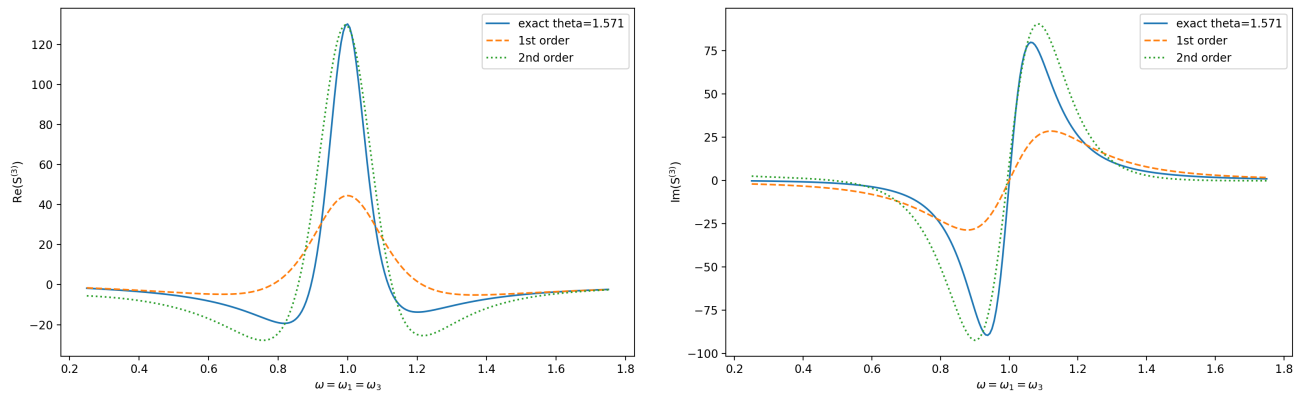


FIG. 3: Comparison between exact and reconstructed signals at maximal misalignment $\theta = \pi/2$. The second-order Duhamel reconstruction reproduces the exact response with high fidelity even in this strongly deformed regime.

Given a reference Liouvillian and an observed spectral response, one seeks the Liouvillian deformation operators required to account for the measured pathway transport. Such a reconstruction would provide a systematic route for identifying missing dynamical processes in approximate open-system models — not by fitting parameters, but by inferring the transport geometry from experimental data. The development of practical reconstruction procedures lies beyond the scope of the present work but is a direct consequence of the framework established here.

VI. DISCUSSION

The results of this work demonstrate that basis incompatibility between the Hamiltonian eigenbasis and the environment-selected pointer basis generates a nontrivial geometric structure in Liouville space — one that leaves measurable signatures in multidimensional spectra and admits a compact description through a low-order transport expansion. We now place these results in broader context.

Multidimensional spectroscopy thereby becomes a problem of transport in pathway space rather than solely one of pathway assignment. The Liouvillian connection provides a local description of that transport, while the associated curvature quantifies the extent to which competing transport processes fail to commute. Geometry enters not as an external mathematical construction but as a consequence of comparing pathway bases transported through open-system dynamics. It reflects the incompatibility of the coherent, optical, and dissipative sectors of the Liouvillian: when the observational basis coincides with the bath-selected pointer basis, the conventional pathway picture is recovered and no nontrivial transport occurs. Curvature emerges when these structures become misaligned, continuously redistributing amplitude throughout the pathway network and generating transport between pathways that would otherwise remain

distinct.

The minimal model considered here isolates this mechanism in its simplest form. A controlled mismatch between the observational basis and the pointer basis generates pathway transport, spectral distortions, and a non-trivial observational holonomy while remaining simple enough to permit direct comparison with the perturbative reconstruction. Its purpose is not to represent a particular physical environment but to isolate the essential ingredients of transport geometry in their simplest form. The underlying geometric structure depends only on the relative orientation of the coherent and dissipative sectors of the Liouvillian, not on the specific choice of representation. One may equivalently hold the dissipative operator fixed and rotate the Hamiltonian through a coherent mixing term; the two descriptions are related by a unitary transformation and generate the same transport geometry. In this sense the geometric response is gauge invariant: it reflects an intrinsic property of the Liouvillian transport network.

We emphasize that the present framework is formulated entirely at the level of the reduced open quantum system. At the level of the combined system and environment, the total dynamics are unitary and may always be represented as coherent evolution on an enlarged Hilbert space. These descriptions are not contradictory. Rather, the reduced Liouvillian, obtained after tracing over the environmental degrees of freedom, provides the effective dynamical generator directly interrogated by nonlinear spectroscopic measurements and therefore defines the observational pathway structure considered throughout this work.

The objective of the present framework is therefore not to reconstruct the microscopic bath dynamics themselves, but to characterize the transport structure induced in the effective reduced Liouvillian. The relevant geometric objects are not the microscopic bath modes, spectral densities, or memory kernels, but the deformations those mechanisms induce in the reduced Liouvillian and, consequently, in the network of observational path-

ways.

This viewpoint differs in spirit from conventional microscopic formulations of open quantum dynamics. Standard approaches begin with a detailed model of the system, bath, and their interaction, deriving an effective equation of motion through approximations leading to Redfield, Nakajima–Zwanzig, or Lindblad descriptions. The goal is to account for observable dynamics by making the microscopic model increasingly complete. The present framework asks a different question. Rather than reconstructing every microscopic environmental degree of freedom, we identify the transport structure required to account for the observed spectroscopic response. The relevant objects are not bath modes, spectral densities, or memory kernels, but the deformations those mechanisms induce on the network of observational pathways. In this respect the framework is closer in spirit to Lindblad’s original formulation — which characterizes the most general physically admissible evolution consistent with structural principles — than to microscopic derivations that require a specific bath realization.

The spectral distortions produced by basis misalignment are not merely signatures of relaxation or environmental broadening. Peak shifts, intensity redistribution, and the appearance of otherwise-forbidden pathway contributions are observable consequences of transport through a curved pathway manifold — geometric signatures as direct as the phase shift produced by a closed loop in parameter space.

The Duhamel expansion provides a practical means of extracting this transport information from spectroscopic data. By expressing the response as systematic deformations of a reference Liouvillian, the expansion identifies the transport operators responsible for the observed redistribution of pathway weight. Rather than introducing increasingly complex phenomenological corrections, it decomposes discrepancies between the reference and observed spectra into physically meaningful transport processes acting on the pathway network. The objective shifts from improving the fit to identifying the transport mechanisms responsible for the observed deformation. In this way, discrepancies between theory and experiment become a source of geometric information rather than simply evidence of an incomplete model. The resulting hierarchy of corrections remains tied to observable pathway transport and therefore preserves physical interpretability as model complexity increases.

This perspective suggests a broader program of geometric inverse spectroscopy: rather than seeking a complete microscopic reconstruction of the environment, one seeks to reconstruct the transport geometry it induces. The resulting description is necessarily coarser than a full microscopic theory, but it is closer to observation. What is universal is not the specific dissipative mechanism but the transport structure it imposes on the pathway network. The geometric description applies whenever environmental interactions redistribute amplitude among observational pathways in competition with coherent evo-

lution — independent of the microscopic origin of that competition.

An important consequence of this viewpoint is that the information contained in a multidimensional spectrum depends not only on the underlying Liouvillian transport but also on the observables used to interrogate it³¹. Different measurement channels may reveal complementary aspects of the same transport geometry while obscuring others. The inverse problem is therefore not simply the reconstruction of dynamics from spectra, but the reconstruction of dynamics from observable projections of those spectra. Understanding this relationship may provide a route toward a broader theory of observability in open quantum systems, linking Liouvillian transport, spectroscopic measurement, and dynamical inference within a common geometric framework.

The geometric structures identified here differ fundamentally from those of equilibrium thermodynamics, quantum-state geometry, or adiabatic phase theory. They emerge not from equilibrium state manifolds, state distinguishability, or geometric phase accumulation, but from the flow of information through open-system pathway space. Dissipation is no longer merely a mechanism for destroying coherence; it becomes a generator of geometric transport. Geometry resides neither in the quantum state alone nor in the environment alone, but in the transport network linking observational pathways.

Multidimensional spectra, viewed from this perspective, carry geometric signatures of the transport structure governing open quantum dynamics. The resulting observational holonomies are experimentally accessible and point toward a broader framework in which coherence, dissipation, and measurement are understood as manifestations of a common transport geometry. Pathway geometry represents a geometric layer of organization in nonlinear spectroscopy complementary to energetic, dynamical, and statistical descriptions — one that reveals aspects of quantum dynamics inaccessible from the state, the environment, or the spectrum considered in isolation.

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Use of Generative Artificial Intelligence

In compliance with institutional guidelines of the Université de Montréal, generative artificial intelligence tools were used to assist with the editing of language and stylistic refinement of parts of the manuscript and to assist in the synthesis of the literature. These tools were not used to generate scientific content, perform analysis, or influence the interpretation of results. All content has been reviewed and validated by the authors, who assume full responsibility for the manuscript.

DATA AVAILABILITY

The numerical data and code that support the findings of this study are openly available in the `Duhamel.Transport` repository, finite-difference release branch `codex/jcp3-finite-difference`. The archived release is deposited on Zenodo with DOI 10.5281/zenodo.21793116. This archive contains the finite-difference two-level-system plus rotated-dephasing implementation, scripts for regenerating the validation calculations and plots, and compact numerical output files used for the pathway-transport diagnostics.

Appendix A: Axiomatic Foundations of Liouvillian Transport and Observational Holonomy

The theory developed in this work is founded upon a small set of geometric principles that distinguish temporal evolution from the quasi-static transport of physical models. The purpose of this appendix is to define the primitive mathematical objects underlying Liouvillian transport and to place the concept of observational holonomy on a precise operational footing.

Throughout, the emphasis is on the minimal geometric structure required to describe nonlinear spectroscopic transport. A more general formulation based on state-generator manifolds and induced response geometry lies beyond the scope of the present work and will be presented elsewhere.

1. Temporal Evolution versus Parametric Transport

The central construction of this work requires distinguishing two fundamentally different notions of evolution.

The first is the familiar temporal evolution of an open quantum system,

$$\frac{d\rho}{dt} = \mathcal{L}(\lambda)\rho, \quad (\text{A1})$$

generated by the Liouvillian with respect to laboratory time t .

The second concerns comparisons between neighboring physical models whose Liouvillians differ infinitesimally through changes of externally controlled parameters,

$$\mathcal{L}(\lambda) \longrightarrow \mathcal{L}(\lambda + d\lambda). \quad (\text{A2})$$

These parameter variations define a quasi-static transport on the manifold of physical models. Here “quasi-static” is used in the thermodynamic sense: neighboring points correspond to neighboring physical models connected through infinitesimal parameter changes. No assumption is made regarding slow temporal evolution, and the present construction is unrelated to the quantum adiabatic theorem.

Accordingly,

$$d\lambda^\mu \neq dt, \quad (\text{A3})$$

and the geometric transport developed below acts with respect to parameter displacements rather than laboratory time.

Axiom I (Manifold of Physical Models)

The collection of admissible open-system models forms a differentiable manifold

$$\mathcal{M} = \{\mathcal{L}(\lambda)\}, \quad (\text{A4})$$

whose coordinates are the externally controlled parameters λ . Each point of $\mathcal{M}$ represents a complete physical model rather than an instantaneous dynamical state.

Axiom II (Quasi-Static Parametric Transport)

A smooth curve

$$\gamma : [0, 1] \rightarrow \mathcal{M} \quad (\text{A5})$$

defines a continuous family of neighboring physical models. Transport along γ compares neighboring Liouvillian generators,

$$\mathcal{L}(\lambda) \rightarrow \mathcal{L}(\lambda + d\lambda), \quad (\text{A6})$$

without implying temporal evolution of a single model. The transport parameter labels position on the model manifold and possesses no independent dynamical significance.

Axiom III (Observable Pathway Bundle)

Associated with every point

$$\lambda \in \mathcal{M} \quad (\text{A7})$$

is a vector space

$$\mathcal{P}_\lambda, \quad (\text{A8})$$

whose elements represent the Liouville-space pathways accessible to a specified nonlinear spectroscopic experiment.

The collection

$$\mathcal{P} = \bigcup_{\lambda \in \mathcal{M}} \mathcal{P}_\lambda \quad (\text{A9})$$

forms the *observable pathway bundle* over the manifold of physical models. No preferred basis is assumed. Different pathway representations correspond to different local bases of the same fiber.

Theorem 1 (Duhamel-Induced Parametric Transport)

Let

$$\mathcal{G}(\xi; \lambda)$$

denote the propagator associated with the Liouvillian $\mathcal{L}(\lambda)$, where ξ represents the propagation variable. Depending on the chosen representation,

$$\mathcal{G}(\xi; \lambda) = \begin{cases} e^{\tau \mathcal{L}(\lambda)}, & \xi = \tau, \\ (i\omega - \mathcal{L}(\lambda))^{-1}, & \xi = \omega. \end{cases}$$

Throughout this appendix, the propagation variable ξ is held fixed while the model parameters λ^μ vary quasi-statically.

Accordingly, the propagator depends upon two independent coordinates,

$$\mathcal{G} = \mathcal{G}(\xi; \lambda),$$

where ξ denotes the propagation variable associated with the chosen representation of the response function and $\lambda \in \mathcal{M}$ labels the underlying physical model.

In the time-domain formulation,

$$\mathcal{G}(\tau; \lambda) = e^{\tau \mathcal{L}(\lambda)},$$

where τ is the physical propagation interval. In the frequency-domain formulation,

$$\mathcal{G}(\omega; \lambda) = (i\omega - \mathcal{L}(\lambda))^{-1},$$

where ω denotes the Fourier conjugate frequency.

Throughout the present construction, differentiation with respect to the model coordinates,

$$\partial_\mu \equiv \frac{\partial}{\partial \lambda^\mu},$$

is performed at fixed propagation variable ξ and generates quasi-static transport on the manifold of physical models.

Consequently, an infinitesimal displacement on the manifold of physical models,

$$\lambda^\mu \rightarrow \lambda^\mu + d\lambda^\mu, \quad (\text{A10})$$

induces the first-order variation

$$\mathcal{G}(\xi; \lambda + d\lambda) = \mathcal{G}(\xi; \lambda) + d\lambda^\mu \partial_\mu \mathcal{G}(\xi; \lambda) + O(d\lambda^2), \quad (\text{A11})$$

where the propagation variable ξ is held fixed. The explicit form of $\partial_\mu \mathcal{G}$ depends upon the representation chosen for the Liouvillian propagator.

The infinitesimal transport generator is defined by

$$K_\mu = (\partial_\mu \mathcal{G}) \mathcal{G}^{-1}, \quad (\text{A12})$$

where $\mathcal{G}$ denotes the chosen representation of the Liouvillian propagator. Explicit expressions for the temporal and Fourier representations follow below.

a. Temporal form For the temporal representation,

$$\mathcal{G}_t(\tau; \lambda) = e^{\tau \mathcal{L}(\lambda)}, \quad (\text{A13})$$

the Duhamel identity gives

$$\partial_\mu \mathcal{G}_t(\tau; \lambda) = \int_0^\tau ds e^{(\tau-s)\mathcal{L}} (\partial_\mu \mathcal{L}) e^{s\mathcal{L}}. \quad (\text{A14})$$

Defining the right transport generator by

$$K_\mu^{(t)} = (\partial_\mu \mathcal{G}_t) \mathcal{G}_t^{-1}, \quad (\text{A15})$$

yields

$$K_\mu^{(t)}(\tau; \lambda) = \int_0^\tau ds e^{(\tau-s)\mathcal{L}} (\partial_\mu \mathcal{L}) e^{-(\tau-s)\mathcal{L}}. \quad (\text{A16})$$

b. Resolvent form. For the frequency-domain (resolvent) propagator,

$$\mathcal{G}_\omega(\omega; \lambda) = [i\omega - \mathcal{L}(\lambda)]^{-1}, \quad (\text{A17})$$

the corresponding parameter derivative follows from the resolvent identity,

$$\partial_\mu \mathcal{G}_\omega(\omega; \lambda) = \mathcal{G}_\omega(\omega; \lambda) (\partial_\mu \mathcal{L}) \mathcal{G}_\omega(\omega; \lambda). \quad (\text{A18})$$

The associated right transport generator is therefore

$$K_\mu^{(\omega)}(\omega; \lambda) = \mathcal{G}_\omega(\omega; \lambda) (\partial_\mu \mathcal{L}) = [i\omega - \mathcal{L}(\lambda)]^{-1} (\partial_\mu \mathcal{L}). \quad (\text{A19})$$

Equations (A16) and (A19) are the temporal and Fourier representations of the same parametric transport construction: in both cases the propagation coordinate is held fixed while the model coordinates λ^μ are varied.

Remark.

It is important to distinguish the Liouvillian deformation,

$$\partial_\mu \mathcal{L}, \quad (\text{A20})$$

from the infinitesimal transport generator,

$$K_\mu, \quad (\text{A21})$$

and the resulting finite transport operator,

$$T_\gamma. \quad (\text{A22})$$

The quantity $\partial_\mu \mathcal{L}$ is the tangent vector describing the infinitesimal deformation of the Liouvillian on the manifold of physical models. The transport generator K_μ is the induced operator-valued one-form that determines how propagators are transported between neighboring models. Integration of this one-form along a quasi-static path $\gamma \subset \mathcal{M}$ produces the finite transport operator,

$$T_\gamma = \mathcal{P} \exp \left(\int_\gamma K_\mu d\lambda^\mu \right). \quad (\text{A23})$$

The hierarchy

$$\partial_\mu \mathcal{L} \longrightarrow K_\mu d\lambda^\mu \longrightarrow T_\gamma \quad (\text{A24})$$

parallels the familiar construction in differential geometry of a connection one-form and its associated parallel transport.

Definition 1 (Finite Parametric Transport)

Let

$$\gamma : [0, 1] \rightarrow \mathcal{M}, \quad s \mapsto \lambda(s), \quad (\text{A25})$$

be a smooth path on the manifold of physical models. The finite transport operator associated with γ is defined by the path-ordered exponential of the Liouvillian transport one-form,

$$T_\gamma = \mathcal{P} \exp \left(\int_\gamma \mathcal{A} \right) = \mathcal{P} \exp \left(\int_\gamma K_\mu d\lambda^\mu \right). \quad (\text{A26})$$

Here $\mathcal{P}$ denotes ordering with respect to the path parameter s , thereby preserving the sequence in which neighboring physical models are traversed along γ . Path ordering is entirely independent of the temporal ordering associated with Liouvillian evolution; it reflects only the ordering of quasi-static displacements on the manifold of physical models.

Axiom IV (Observational Completion)

Transport of pathway amplitudes through the observable pathway bundle defines an operator-valued geometric quantity but does not by itself constitute an experimentally measurable observable. In nonlinear spectroscopy, the transport process is completed by the terminal trace operation, which contracts the transported

pathway with the preparation and detection operators defining the experiment.

Operationally, this completion is the familiar trace appearing in nonlinear response theory,

$$S = \text{Tr} [OT_\gamma(\rho_0)] = \langle\langle O|T_\gamma|\rho_0 \rangle\rangle. \quad (\text{A27})$$

We refer to this contraction as *observational completion*.

Definition 3 (Observational Holonomy)

Transport of pathway amplitudes through the observable pathway bundle defines an operator-valued geometric quantity but does not, by itself, constitute an experimentally measurable observable. In nonlinear spectroscopy, the transport process is completed by the terminal trace operation, which contracts the transported pathway with the preparation and detection operators defining the experiment.

$$\mathcal{H}_{\text{obs}}(\gamma) = \text{Tr} [OT_\gamma(\rho_0)] = \langle\langle O|T_\gamma|\rho_0 \rangle\rangle. \quad (\text{A28})$$

Unlike conventional geometric holonomy, which is associated with parallel transport around a closed contour in parameter space, observational holonomy is obtained through quasi-static transport on the manifold of physical models followed by observational completion through the preparation and detection maps of the spectroscopic experiment.

This construction is operationally analogous to the evaluation of thermodynamic work over a closed cycle. In equilibrium thermodynamics, the infinitesimal work one-form is integrated along a prescribed thermodynamic path to produce a measurable scalar. Here, the Liouvillian transport one-form is integrated to generate the finite transport operator, whose observational completion yields the measurable spectroscopic response.

Theorem 1 (Existence of the Liouvillian Transport One-Form)

For the smooth family of Liouvillian propagators $\mathcal{G}(\xi; \lambda)$ defined over the manifold of physical models $\mathcal{M}$, the infinitesimal parametric variation

$$\lambda^\mu \rightarrow \lambda^\mu + d\lambda^\mu \quad (\text{A29})$$

induces an operator-valued differential one-form

$$\mathcal{A} = K_\mu d\lambda^\mu, \quad (\text{A30})$$

where

$$K_\mu = (\partial_\mu \mathcal{G}) \mathcal{G}^{-1}. \quad (\text{A31})$$

The one-form $\mathcal{A}$ is independent of the choice of coordinates on $\mathcal{M}$ and generates finite parametric transport through its path-ordered exponential.

Proof. Since $\mathcal{G}(\xi; \lambda)$ is differentiable on the smooth manifold $\mathcal{M}$, its first-order variation under an infinitesimal parameter displacement is

$$\mathcal{G}(\xi; \lambda + d\lambda) = \mathcal{G}(\xi; \lambda) + d\lambda^\mu \partial_\mu \mathcal{G}(\xi; \lambda) + O(d\lambda^2). \quad (\text{A32})$$

The explicit form of $\partial_\mu \mathcal{G}$ depends upon the chosen representation of the Liouvillian propagator. For temporal propagation, Eq. (A14) yields the Duhamel representation, whereas for the resolvent form Eq. (A18) follows directly from the resolvent identity. In either representation,

$$K_\mu = (\partial_\mu \mathcal{G}) \mathcal{G}^{-1} \quad (\text{A33})$$

defines the infinitesimal transport generator.

To establish that

$$\mathcal{A} = K_\mu d\lambda^\mu$$

is a differential one-form, consider a smooth change of coordinates

$$\lambda^\mu \rightarrow \lambda^{\mu'}(\lambda). \quad (\text{A34})$$

The parameter derivatives transform according to the chain rule,

$$\partial_{\mu'} = \frac{\partial \lambda^\nu}{\partial \lambda^{\mu'}} \partial_\nu, \quad (\text{A35})$$

so that

$$K_{\mu'} = \frac{\partial \lambda^\nu}{\partial \lambda^{\mu'}} K_\nu. \quad (\text{A36})$$

Similarly,

$$d\lambda^{\mu'} = \frac{\partial \lambda^{\mu'}}{\partial \lambda^\rho} d\lambda^\rho. \quad (\text{A37})$$

Consequently,

$$\begin{aligned} K_{\mu'} d\lambda^{\mu'} &= \left(\frac{\partial \lambda^\nu}{\partial \lambda^{\mu'}} K_\nu \right) \left(\frac{\partial \lambda^{\mu'}}{\partial \lambda^\rho} d\lambda^\rho \right) \\ &= K_\rho d\lambda^\rho, \end{aligned} \quad (\text{A38})$$

where the Jacobians cancel by the chain rule. Hence,

$$\mathcal{A} = K_\mu d\lambda^\mu \quad (\text{A39})$$

is independent of the choice of coordinates and therefore defines an operator-valued differential one-form on the manifold of physical models.

Finally, integration of $\mathcal{A}$ along a smooth path $\gamma \subset \mathcal{M}$ generates the finite transport operator

$$T_\gamma = \mathcal{P} \exp \left(\int_\gamma \mathcal{A} \right), \quad (\text{A40})$$

completing the proof. $\square$

The introduction of the Liouvillian transport one-form is essential because observational holonomy is fundamentally a path-dependent quantity. As in equilibrium thermodynamics, where the infinitesimal work one-form provides the differential object whose integration along a thermodynamic path yields the accumulated work, the transport one-form $\mathcal{A} = K_\mu d\lambda^\mu$ provides the infinitesimal geometric object whose path-ordered integration generates finite Liouvillian transport. Observational holonomy is therefore not defined directly from the propagator itself, but from the observable obtained after observational completion of the finite transport generated by the underlying transport one-form.

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